

Linear-Time Demonstration Selection for In-Context Learning via Gradient Estimation

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Motivation

- In-context learning has emerged as a paradigm for solving and generating natural language tasks using LLMs. We study the problem of selecting a fixed-size subset of demonstration examples given a query set.
- Several works have working on this before.
 - Liu et al., 2022 select the most similar demonstrations based on their input embeddings. But it overlooks the model output conditioned on the demonstration labels.
 - Peng et al. 2024 consider the output loss of different demonstrations, but compute the output is costly when the number of demonstration set is quite large.



Overview

- Formally, given a demonstration set, training set, and query set, how can we select the most valuable data points from the demonstration set that improve the query output most?
- Based on the literature, we propose GradSel.
 - We consider the output of LMs, while we use first-order Taylor expansion to efficiently estimate the inference result.
 - Based on this, we utilize the random ensemble method to select the demonstration data with the best attributions.



Scaling Up Inference via Estimation

- Consider the model output of a prompt subset S on an input query x , denoted as $f_W(\phi(S, x))$.
- Given an anchor prompt S_0 , the first-order approximation of f_W around the embedding vector $\phi(S_0, x)$ is:

$$f_W(\phi(S, x)) = f_W(\phi(S_0, x)) + \langle \nabla_{\phi} f_W(\phi(S_0, x)), \phi(S, x) - \phi(S_0, x) \rangle + \epsilon_{S,x},$$

where ∇_{ϕ} denotes the gradient of f_W with respect to the input embedding ϕ and $\epsilon_{S,x}$ denotes the approximation error.

- To evaluate the accuracy of this approximation, we calculate the relative

$$\text{error} \left\| \frac{\epsilon_{S,x}}{f_W(\phi(S, x))} \right\|^2.$$



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Distance	DeepSeek-7B	Llama-13B	CodeLlama-34B
15% – 20%	0.21 $\pm$ 0.04%	0.06 $\pm$ 0.00%	0.06 $\pm$ 0.01%
20% – 25%	0.30 $\pm$ 0.05%	0.08 $\pm$ 0.02%	0.06 $\pm$ 0.01%
25% – 30%	0.51 $\pm$ 0.04%	0.09 $\pm$ 0.01%	0.08 $\pm$ 0.02%
30% – 35%	0.53 $\pm$ 0.07%	0.28 $\pm$ 0.03%	0.13 $\pm$ 0.03%
35% – 40%	1.40 $\pm$ 0.20%	0.37 $\pm$ 0.03%	0.30 $\pm$ 0.03%

Language Model	Poem-Sentiment	Edge-Existence	CR	Modular-Addition
Llama-3.2-1B	3.7 $\pm$ 0.1 $\times 10^{-3}$	8.9 $\pm$ 0.7 $\times 10^{-4}$	1.8 $\pm$ 0.3 $\times 10^{-2}$	6.1 $\pm$ 1.6 $\times 10^{-3}$
OPT-1.3B	1.4 $\pm$ 0.3 $\times 10^{-2}$	1.1 $\pm$ 0.1 $\times 10^{-2}$	2.5 $\pm$ 0.6 $\times 10^{-2}$	2.3 $\pm$ 1.1 $\times 10^{-3}$
Llama-3.2-3B	1.7 $\pm$ 0.7 $\times 10^{-3}$	6.1 $\pm$ 0.2 $\times 10^{-3}$	3.3 $\pm$ 1.0 $\times 10^{-3}$	3.5 $\pm$ 1.0 $\times 10^{-3}$
DeepSeek-7B	2.8 $\pm$ 0.9 $\times 10^{-3}$	1.6 $\pm$ 0.2 $\times 10^{-3}$	1.7 $\pm$ 0.9 $\times 10^{-2}$	3.2 $\pm$ 0.1 $\times 10^{-3}$
Llama-3.1-8B	4.3 $\pm$ 0.6 $\times 10^{-3}$	1.6 $\pm$ 0.1 $\times 10^{-2}$	1.8 $\pm$ 0.1 $\times 10^{-2}$	4.2 $\pm$ 1.2 $\times 10^{-3}$
Llama-2-13B	1.0 $\pm$ 0.6 $\times 10^{-3}$	1.4 $\pm$ 0.0 $\times 10^{-3}$	1.6 $\pm$ 0.4 $\times 10^{-4}$	8.3 $\pm$ 1.6 $\times 10^{-3}$
CodeLlama-34B	4.9 $\pm$ 2.9 $\times 10^{-4}$	6.4 $\pm$ 1.0 $\times 10^{-4}$	1.0 $\pm$ 0.2 $\times 10^{-3}$	1.6 $\pm$ 0.2 $\times 10^{-3}$



Efficient Demonstration Selection

- We propose the random ensemble method called ICL-GradRE, which is inspired by the literature on influence functions (Koh and Liang, 2017) and the work of task affinity grouping (Fifty et al., 2021).
- We draw m random subsets from demonstration set $\{\theta_1, \theta_2, \dots, \theta_n\}$, each of fixed size k , then we estimate f_W for each of the subsets.
- We compute an attribution score s_i for a demonstration θ_i that equals the average of f_W over subsets include θ_i .
- Example:
 - $f_W(\theta_1, \theta_2) = 0.1, f_W(\theta_1, \theta_3) = 0.2, f_W(\theta_2, \theta_3) = 0.3,$
 - Then, $s_1 = \frac{0.1+0.2}{2}, s_2 = \frac{0.1+0.3}{2}, s_3 = \frac{0.2+0.3}{2}.$



A case study of linear functions

- We use an example of learning linear functions with transformer models (Grag et al., 2022).
- Suppose that both the in-context example and query example are generated as $y_i = \langle \beta^{(j)}, x_i \rangle + \epsilon_i$.
- The input vector x_i is randomly sampled from $N(0, Id_d)$ with $d = 20$. The coefficient vector $\beta^{(j)}$ is randomly drawn from a finite set $\{\beta^{(1)}, \beta^{(2)}, \dots, \beta^{(C)}\}$, $C < d$. Different in-context examples may have different coefficients.



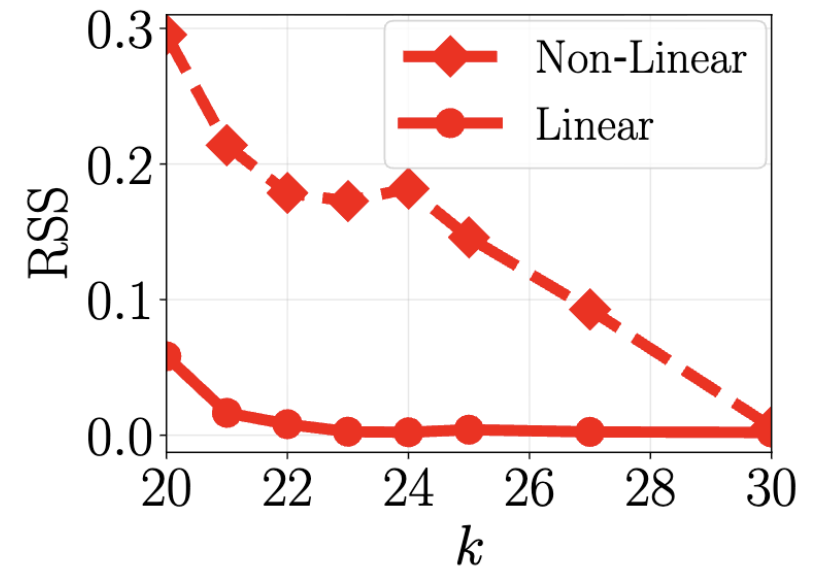
A case study of linear functions

- We use an example of learning linear functions with transformer models (Grag et al., 2022).
- Suppose that both the in-context example and query example are generated as $y_i = \langle \beta^{(j)}, x_i \rangle + \epsilon_i$.
- We consider two settings for the noise term ϵ_i
 - Linear regression with $\epsilon_i = 0$ for all i .
 - Noisy linear regression with $\epsilon_i \sim N(0,1)$.



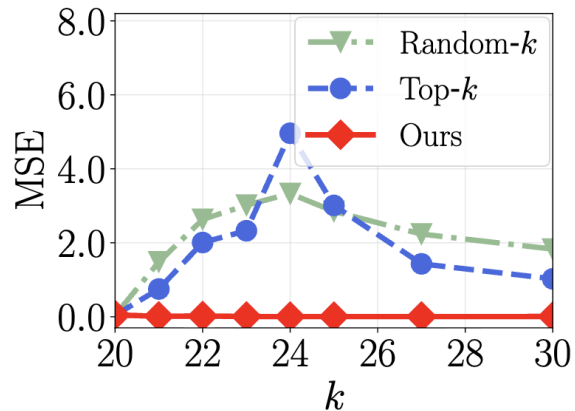
A case study of linear functions

- We first test the approximation accuracy. In both settings, our method can accurately estimate the true outcomes of ICL.
- As k increases, the trained transformer learns the linear function correctly, and the error decreases to near 0.
- Since $d = 20$, we plot from $k = d$ onward because this is the minimum number of samples required to solve a system of linear equations.



A case study of linear functions

- Then, we test the random ensemble method.
- To further evaluate the random ensemble mechanism, we report the average score of demonstration samples from $\beta^{(1)}$ and $\beta^{(2)}$.
 - The query set is generated from $\beta^{(1)}$.
 - We vary the number of subsets and find that the in-distribution samples always achieve lower loss than OOD samples.



# Subsets	5	50	100	1000
Samples from $\beta^{(1)}$	2.1 ± 2.0	3.1 ± 1.1	3.9 ± 0.8	3.7 ± 0.4
Samples from $\beta^{(2)}$	2.6 ± 3.6	4.6 ± 2.7	5.2 ± 2.5	5.4 ± 2.7



Extension

- Theoretically, our estimation method can be utilized to other inference-based selection method.
- We adapt it to forward selection (ICL-GradFS), and cross-entropy based selection (ICL-GradCE).



Experiment setup

- Datasets:
 - Sentiment classification: Poem Sentiment, SST-2, and CR.
 - Reasoning: GraphQA, Modular Addition, Coin-flip.
- Models:
 - Llama-3.2-1B, OPT-1.3B, Llama-3.2-3B, DeepSeek-7B, Qwen-7B, Llama-3.1-8B, Llama-2-13B, CodeLlama-34B.
- Baselines:
 - BM25, Random-k, Top-k, Top-k+CE ([ACL 2024](#)), Uncertainty Ranking ([NAACL 2025](#)), BRIDGE ([ICLR 2025](#)).



Experiment results

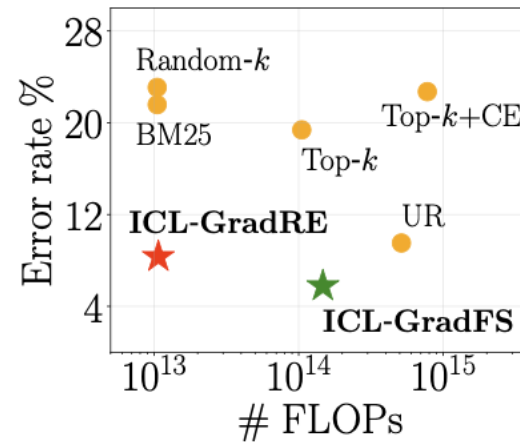
- Regarding the estimation and efficiency, we report the relative error between the estimated loss $\hat{f}_W$ and the true loss f_W .
- To measure computational cost, we count the total number of floating-point operations (FLOPs) performed on an Nvidia A6000 GPU card.

Coin-Flip	DeepSeek-7B	CodeLlama-34B	Speedup
ICL-GRADFS	$0.17 \pm 0.0\%$	$0.52 \pm 0.0\%$	37.7×
ICL-GRADRE	$0.08 \pm 0.1\%$	$0.07 \pm 0.0\%$	19.7×
ICL-GRADCE	$0.58 \pm 0.0\%$	$0.46 \pm 0.0\%$	17.3×
SST-2	DeepSeek-7B	CodeLlama-34B	Speedup
ICL-GRADFS	$0.20 \pm 0.1\%$	$0.43 \pm 0.1\%$	19.0×
ICL-GRADRE	$0.18 \pm 0.0\%$	$0.36 \pm 0.0\%$	10.8×
ICL-GRADCE	$0.80 \pm 0.0\%$	$0.79 \pm 0.0\%$	6.7×

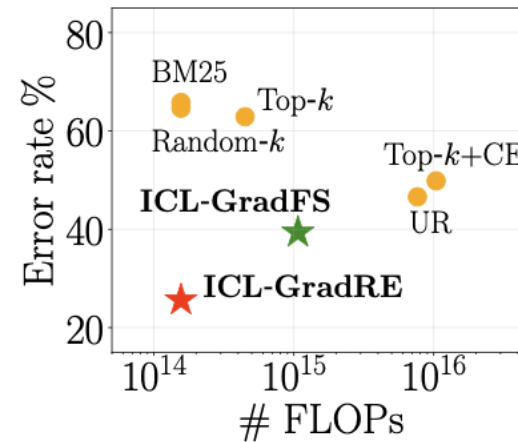


Experiment results

- Regarding the performance of our method, we report the trade off between error rate and FLOPs.
- We use DeepSeek-7B model to test on SST-2 and Coin Flip dataset.



(a) SST-2

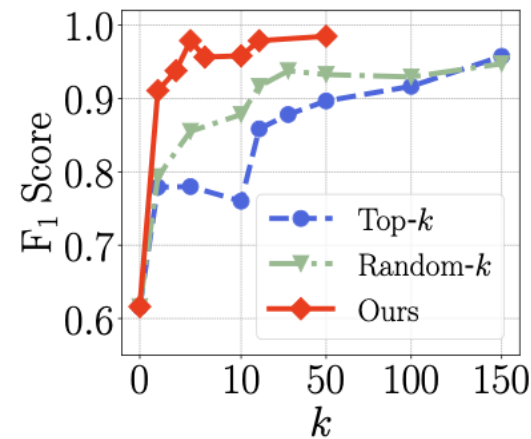


(b) Coin Flip

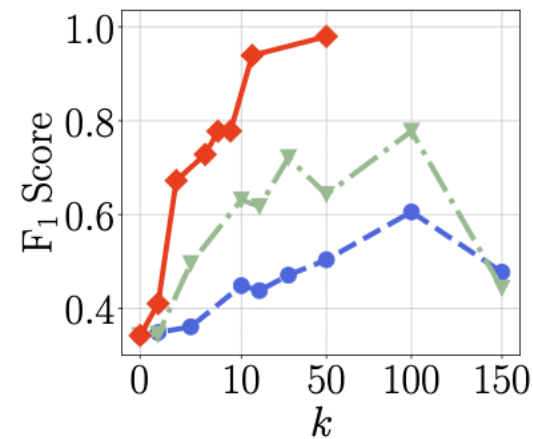


Experiment results

- Regarding the long context setting, we varies the number of selected demonstrations from 0 to 150.
- Compared to Top-k and Random-k, our approach matches 15 less tokens to reach the same performance.



(a) SST-2



(b) Coin Flip

Summary

- We first utilize a first-order approximation to estimate the output of LLM.
- Then, we use a random ensemble method to compute the data attribution score.
- We test our method on simple functions and real world datasets.

Thanks for Listening!

- Paper: <https://arxiv.org/pdf/2508.19999>
- Code: <https://github.com/VirtuosoResearch/ICL-GradSel>
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